

Lineárny operátor na priestore

$$V = \text{Span} \left\{ \underbrace{\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}}_{v_2} \right\}$$

$$f \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}$$

$$f \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

Späčtejte $f^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}$ a zapíšte výsledok vzorcom v závislosti na n .

$$f \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}, \quad f \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}, \quad f \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = v_1 + 3v_2$$

$$f(v_1) = v_1 + 2v_2, \quad f(v_2) = 2v_1 - 2v_2$$

Nechť $f = f_A$, kde je určená maticí A :

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = (1-\lambda) \cdot (-2-\lambda) - 4 \\ &= \lambda^2 + \lambda - 6 \\ &= (\lambda - 2)(\lambda + 3) = 0 \end{aligned}$$

$$\lambda_1 = 2, \quad \lambda_2 = -3$$

$$\lambda_1 = 2 :$$

$$\left(\begin{array}{cc|c} -1 & 2 & 0 \\ 2 & -4 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \rightarrow x_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\lambda_2 = -3 :$$

$$\left(\begin{array}{cc|c} 4 & 2 & 0 \\ 2 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right) \rightarrow x_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Tznize

$$R = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$D = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

a tedy

$$\begin{aligned} A^k &= R D^k \cdot R^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \cdot \begin{pmatrix} 2^k & 0 \\ 0 & (-3)^k \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \cdot \begin{pmatrix} 2^k & 0 \\ 0 & (-3)^k \end{pmatrix} \cdot \begin{pmatrix} 2/5 & 1/5 \\ 1/5 & -2/5 \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} 2^{k+1} & (-3)^k \\ 2^k & -2 \cdot (-3)^k \end{pmatrix} \cdot \begin{pmatrix} 2/5 & 1/5 \\ 1/5 & -2/5 \end{pmatrix}$$

$$A^k = \frac{1}{5} \cdot \begin{pmatrix} 2^{k+2} + (-3)^k & 2^{k+1} - 2 \cdot (-3)^k \\ 2^{k+1} - 2 \cdot (-3)^k & 2^k + 4 \cdot (-3)^k \end{pmatrix}$$

$$A^k \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \frac{1}{5} \cdot \begin{pmatrix} 2^{k+2} + (-3)^k + 3 \cdot 2^{k+1} - 6 \cdot (-3)^k \\ 2^{k+1} - 2 \cdot (-3)^k + 3 \cdot 2^k + 12 \cdot (-3)^k \end{pmatrix}$$

$$= \frac{1}{5} \cdot \begin{pmatrix} 5 \cdot 2^{k+1} - 5 \cdot (-3)^k \\ 5 \cdot 2^k + 10 \cdot (-3)^k \end{pmatrix}$$

$$A^k \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2^{k+1} - (-3)^k \\ 2^k + 2 \cdot (-3)^k \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

tanze

$$f^y \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = a \cdot v_1 + b \cdot v_2$$

$$f^y \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = a \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + b \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

$$f^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 2a+b \\ a \\ b \\ a+2b \end{pmatrix}$$

$$f^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 2 \cdot (2^{k+1} - (-3)^k) + 2^k + 2(-3)^k \\ 2^{k+1} - (-3)^k \\ 2^k + 2(-3)^k \\ 2^{k+1} - (-3)^k + 2(2^k + 2(-3)^k) \end{pmatrix}$$

$$f^n \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^k \\ 2^{k+1} - (-3)^k \\ 2^k + 2(-3)^k \\ 2^{k+2} + 3 \cdot (-3)^k \end{pmatrix}$$

